Assignment 4 – Clustering with a Hugging Face Dataset

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Dataset Description

For this assignment, I choose to use the **AG News** dataset from the Hugging Face Datasets Library. This dataset contains short news articles collected from various sources and represents a realistic natural language processing task. A random subset of **5,000 samples** was selected from the training split to reduce computational cost while maintaining topic diversity. Only the article text was used; original labels were intentionally ignored to preserve the unsupervised nature of the clustering task. The text data was converted into numerical features using **TF-IDF vectorization**, producing a sparse representation suitable for clustering.

Clustering Analysis

K-Means clustering was applied to the TF-IDF feature matrix. To determine the appropriate number of clusters, the **Elbow Method** was used by plotting inertia values for k ranging from 2 to 10. A clear elbow was observed at **k = 5**, indicating diminishing returns beyond this point. The final K-Means model was trained with five clusters. To visualize the clustering results, **Principal Component Analysis (PCA)** was applied to reduce the feature space to two dimensions. The resulting scatter plot showed reasonably well-separated clusters, suggesting meaningful structure in the data. Inspection of the most influential TF-IDF terms per cluster revealed that the clusters corresponded to interpretable news topics such as world affairs, business, sports, and technology.

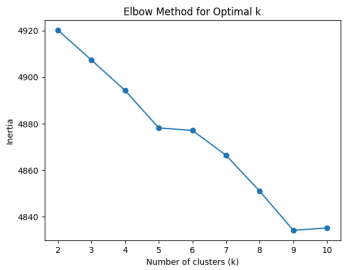

Figure 1 – Elbow Method for Cluster Selection

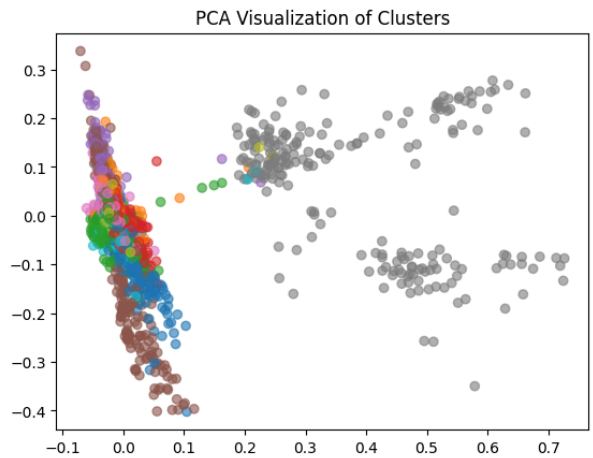

Figure 2 – PCA Visualization of K-Means Clusters

Cluster Prediction with a Supervised Model

After assigning each data point to a cluster, the cluster labels were treated as targets for a supervised learning task. The dataset was split into training and testing sets using an 80/20 split with stratification to preserve cluster proportions. A **Logistic Regression** classifier was trained to predict the cluster assignments based on the TF-IDF features. Model evaluation on the test set showed high predictive performance, with strong accuracy and balanced precision and recall across clusters. This result indicates that the discovered clusters are internally consistent and can be learned effectively by a supervised model.

	precision	recall	f1-score	support
0	1.00	0.68	0.81	66
1	1.00	0.63	0.78	41
2	0.76	1.00	0.86	260
3	0.88	0.90	0.89	172
4	1.00	0.87	0.93	31
5	0.87	0.84	0.86	131
6	0.93	0.89	0.91	163
7	1.00	0.95	0.98	44
8	1.00	0.50	0.67	30
9	1.00	0.74	0.85	62
accuracy			0.87	1000
macro avg	0.94	0.80	0.85	1000
weighted avg	0.89	0.87	0.87	1000

Interpretation and Significance

The clustering results demonstrate that unsupervised learning can successfully uncover latent semantic structure in text data without access to ground-truth labels. The ability of a supervised model to accurately predict cluster assignments further validates the coherence and stability of the clusters. This experiment highlights how clustering can be used for exploratory data analysis, topic discovery, and as a preprocessing step for downstream supervised learning tasks in real-world machine learning workflows.